

# Planespotting from Space:

## Flying Aircraft Detection in Satellite Imagery via Inter-Band Parallax

Luca-Verona Vellage · MDS 2026

### Objectives

This study developed an end-to-end pattern-matching pipeline for flying aircraft detection in Sentinel-2 imagery. Independent monitoring of untracked aircraft is a use case of growing relevance to open-source and investigative research. Detection performance was evaluated across background strata to assess applicability in this context.

### Sentinel-2 Satellite Data

Earth Observation (EO) satellites like Sentinel-2 capture the Earth's surface by recording spectral bands sequentially at fixed time offsets. Any ground point is therefore observed at a slightly different instant per band. Fast-moving objects like aircraft traverse sufficient distance to appear displaced across bands, producing a characteristic parallax signature that forms the basis for detection (see 1).

### Methodology

The hybrid detection framework, adapted from Liu et. al. (2020), combines cloud-based and local computation to enable identification of sparsely distributed aircraft across large geographic areas (see 2).

**Step 1** identifies candidate pixels via reflectance anomaly segmentation in Google Earth Engine (GEE) for any date range and area of interest.

**Step 2** processes binary candidate masks locally to filter the three most common false positive sources.

**Step 3** downloads image chips of surviving candidates from GEE.

**Step 4** confirms aircraft locally based on three physical criteria: Reflectance anomaly relative to local background, collinearity of band anomalies and proportional inter-band displacement (see examples in 2).

**Step 5** cross-references detections against conventional tracking records (ADS-B) and performs visual inspection.

29 scenes containing 301 visually detectable aircraft were evaluated and visually confirmed across five background strata: water, vegetation, barren land, urban and snow.

### Results

Detection performance varied markedly across background strata, with F1-scores ranging from 0.84 over water to 0.04 over snow. Performance increased over dark, homogenous surfaces and degraded over bright and heterogeneous backgrounds, where contrast-based segmentation failed as background reflectance exceeded that of aircraft (see 3).

Beyond ADS-B matched detections, the pipeline identified aircraft without a conventional tracking record, which are not included in the performance metrics. The fraction of such untracked aircraft was highest over vegetation and barren land, coinciding with regions of typically sparse conventional tracking coverage.

### 1 Parallax Effect

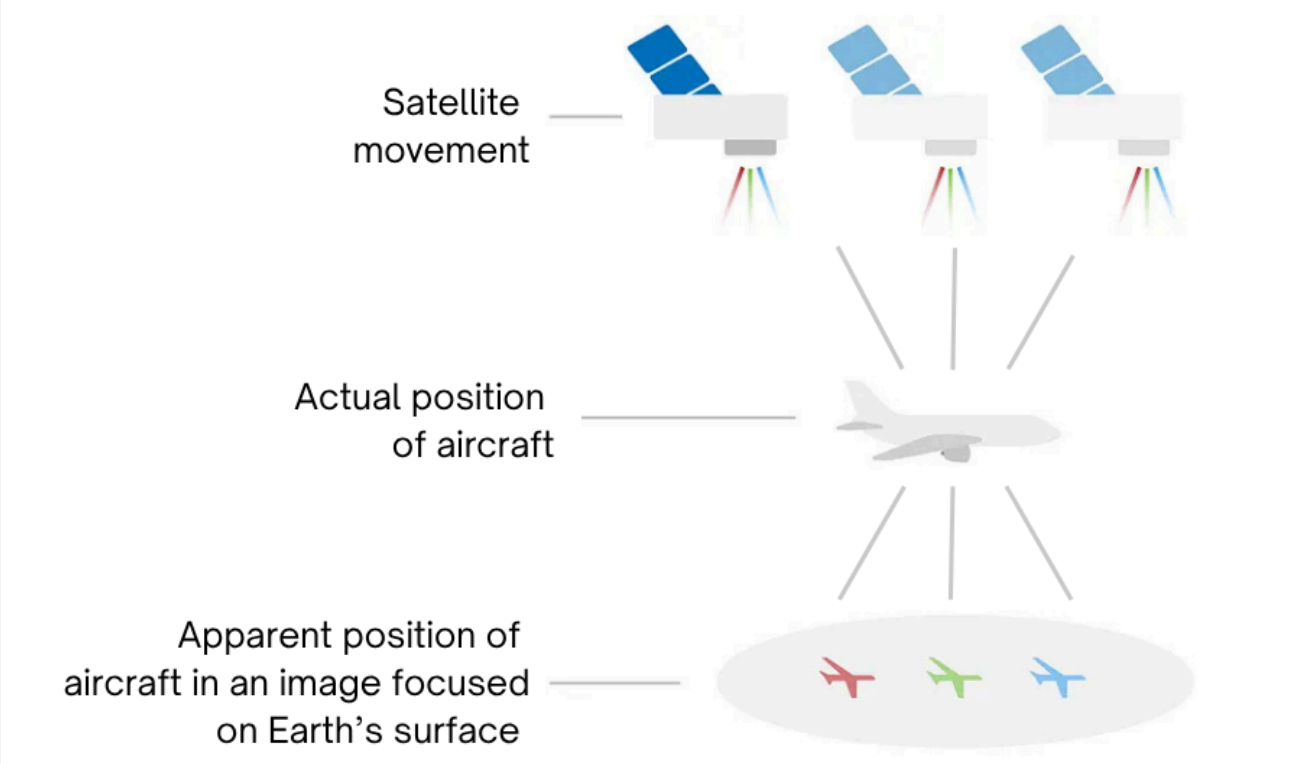


Figure 1: Illustration of spectral band parallax through Sentinel-2 pushbroom motion (Erickson, 2017).

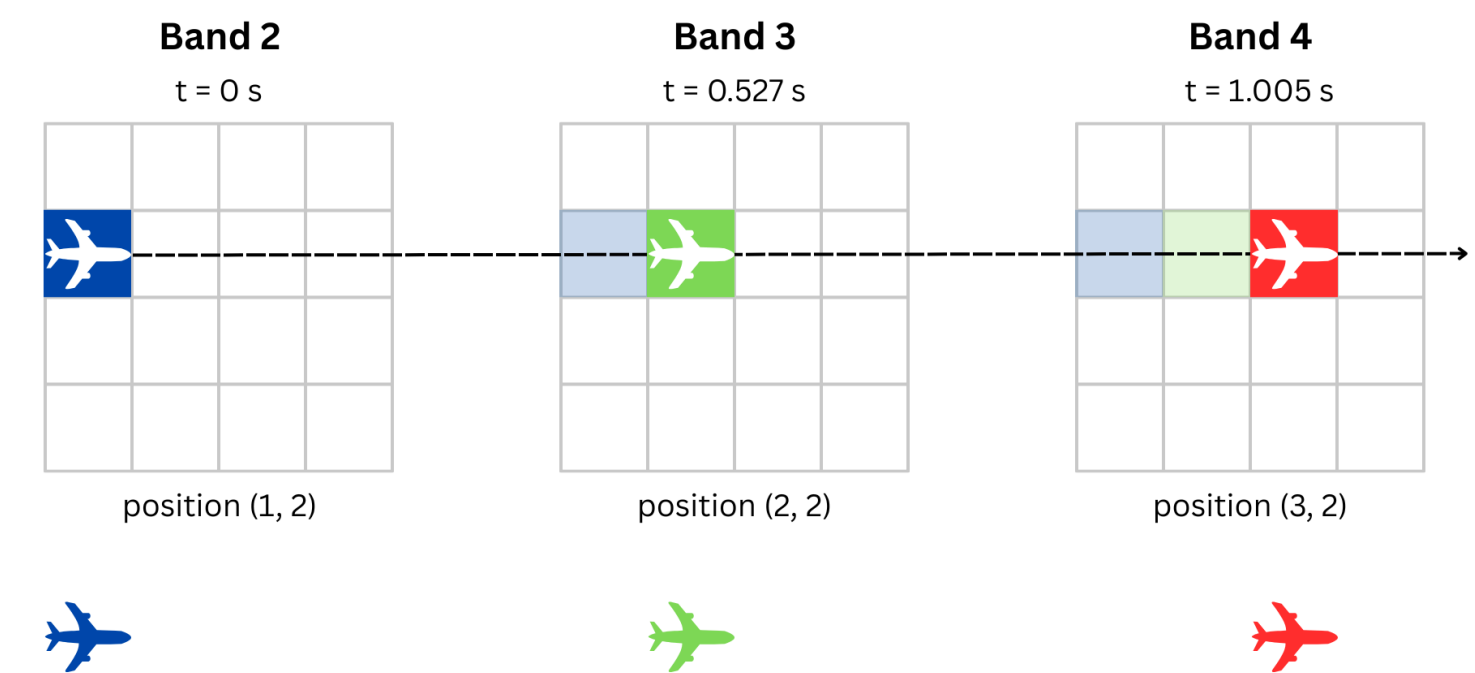
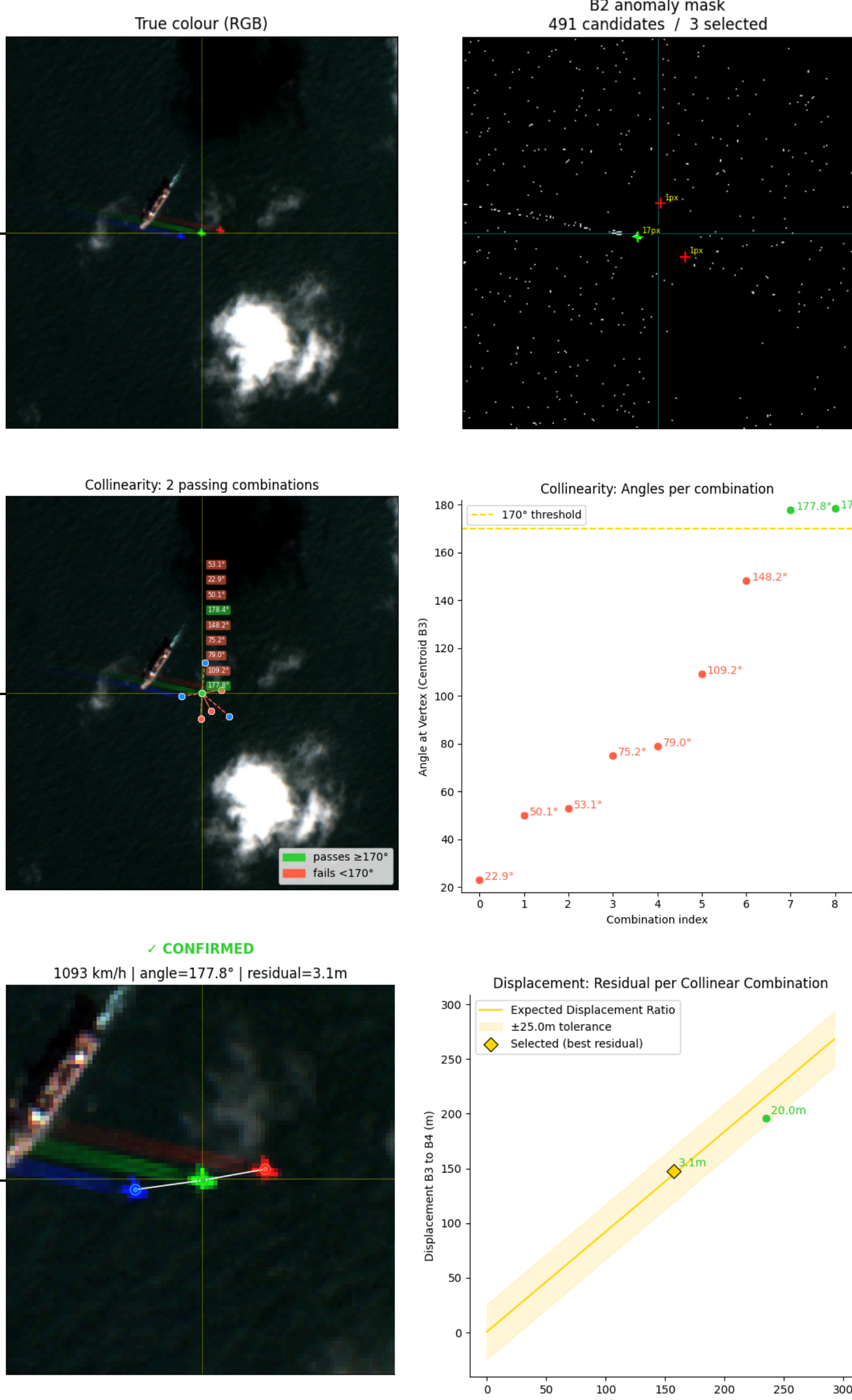
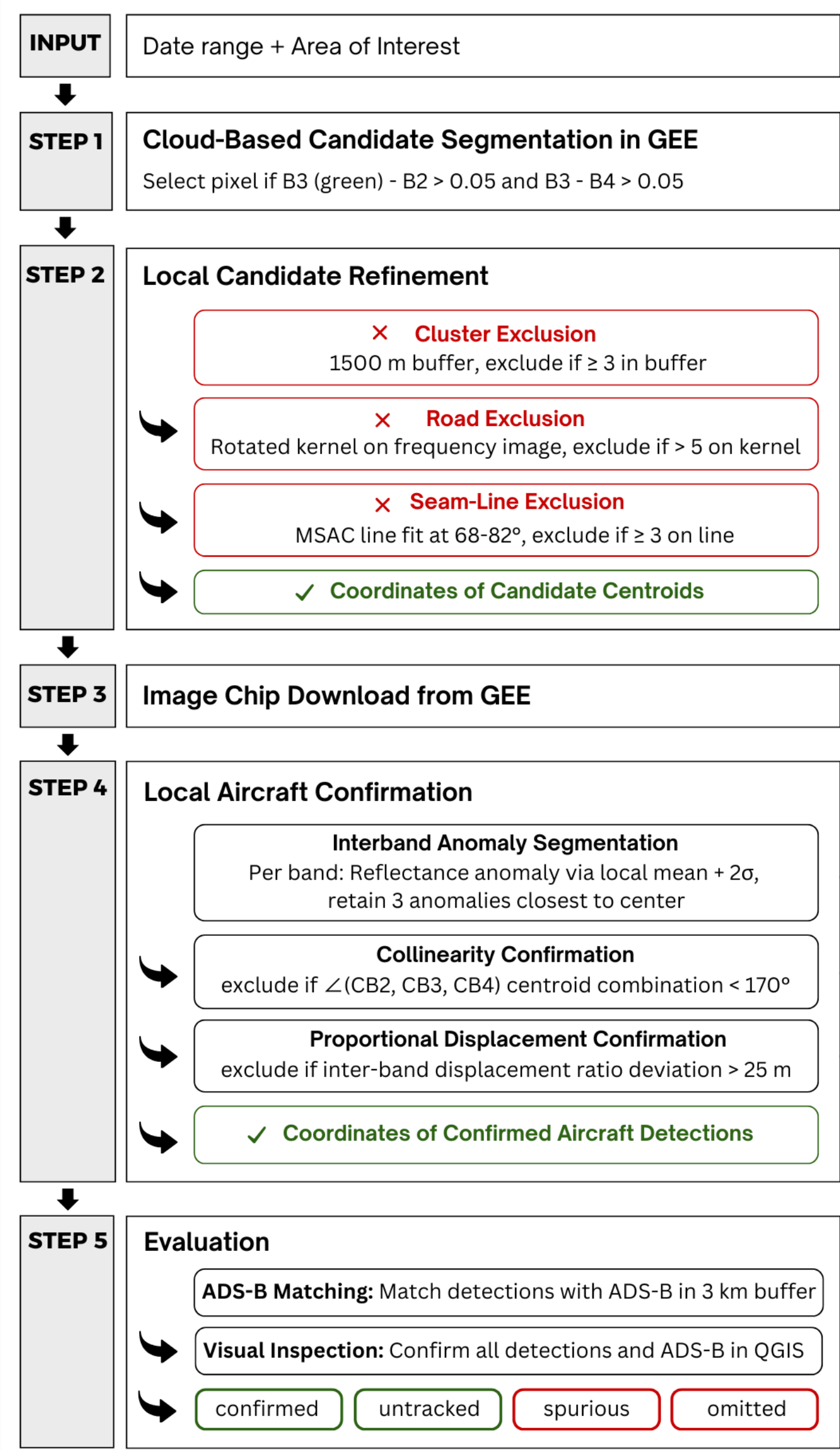


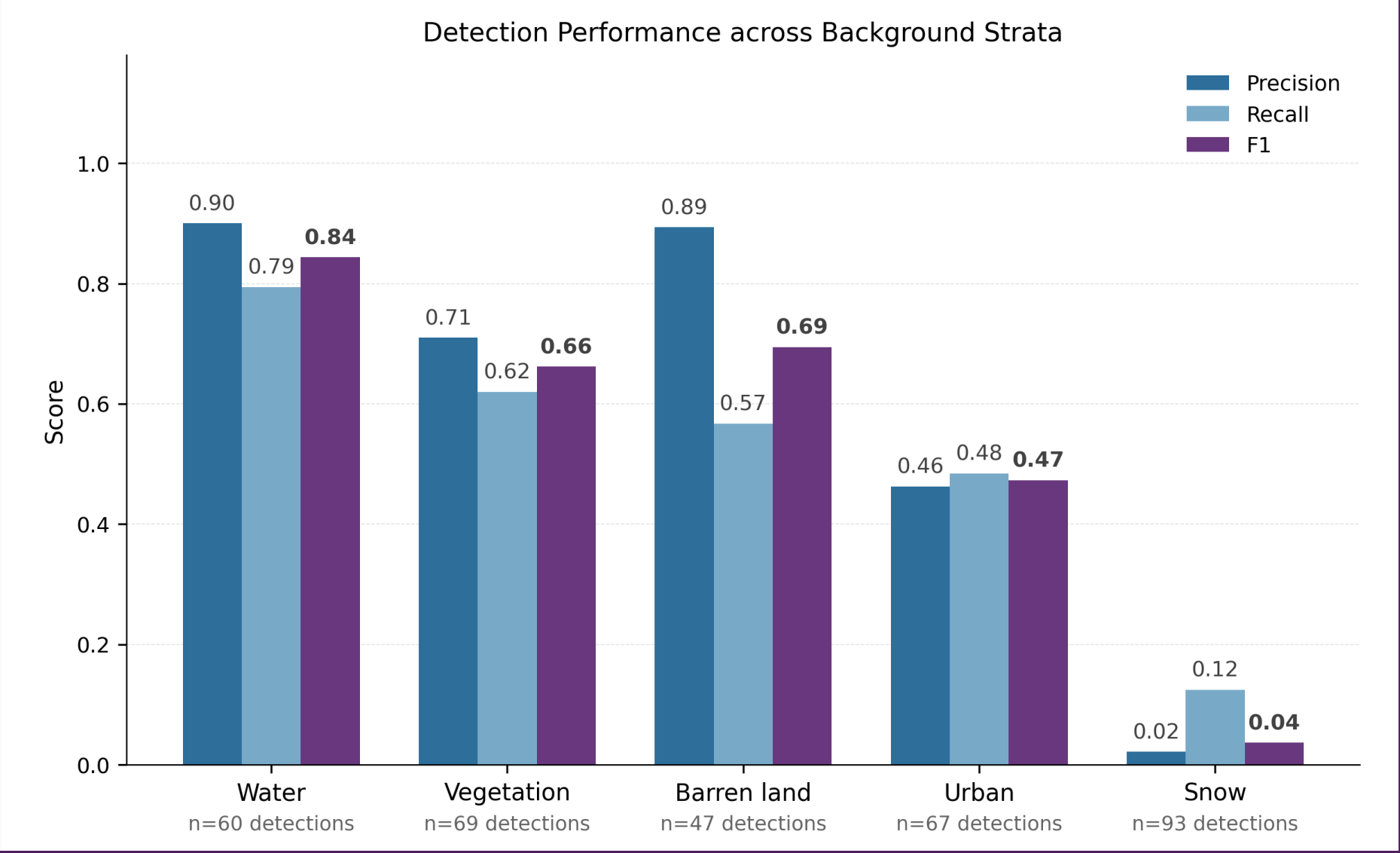
Figure 2: Illustration of the parallax acquisition from the Sentinel-2 perspective.

### 2 Hybrid Detection Pipeline



Figures 3-6: Flowchart of the End-to-End Detection Pipeline with an example of aircraft confirmation in a Sentinel-2 image over the English Channel.

### 3 Evaluation Results



### Policy Implications

Satellite-based aircraft detection constitutes a viable complement to conventional tracking for independent investigation of aircraft movements.

Open Earth Observation data enables civilian-accessible airspace monitoring beyond the reach of state and private actors.

Maintaining open satellite data access is critical for accountability, investigative journalism, and open-source intelligence.